Pandelis Zembashis

CTO | AI Deployment Engineer | Startup AI Systems & Evaluation Loops

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EXECUTIVE SUMMARY

CTO and hands-on AI engineer with deep startup operating experience, spanning agent systems, evaluation loops, customer deployment, product judgment, and engineering leadership. At Cosine, built and led much of the platform behind AI coding agents: prompts and tool contracts, context handling, model/provider behavior, RL-style feedback loops, 20,000+ coding gyms, deterministic replay, secure execution, and enterprise deployment. Strong fit for OpenAI's AI Deployment Engineer - Startups role through direct experience turning messy real-world AI workflows into reproducible evals, product improvements, and better model behavior while working closely with technical users and founders.

PROFESSIONAL EXPERIENCE

Cosine | CTO | Oct. 2023 – May 2026

Applied AI lab building coding agents, model feedback loops, secure execution, and enterprise deployment infrastructure.

- Led product and engineering for Cosine's AI coding platform, working across agent architecture, prompts, tool use, context management, model behavior, customer deployment, reliability, and developer experience.
- Built the core agent harness for streaming tool calls, event scheduling, checkpoints, deterministic replay, memory writes, MCP/LSP integrations, subagents, file locks, repository search, browser use, and structured rollout traces.
- Built ML feedback loops around frontier closed-model agents, including RL-style training/evaluation signals from traces, tests, LSP diagnostics, reviews, conversation patterns, and parallel sandboxed runs.
- Led a multi-month rollout of 20,000+ coding gyms, turning real agent attempts into repeatable evaluation environments for improving tool use, planning, recovery, and long-horizon code modification.
- Worked across OpenAI, Anthropic, Gemini, and internal model capabilities, shaping prompts, reward/evaluation signals, tool contracts, context strategy, and provider-specific behavior into stable product patterns.
- Built deployment paths for managed cloud, private VPC, on-prem, and air-gapped customers where code, logs, model artifacts, and tenant data needed to remain isolated, observable, and auditable.
- Turned customer and internal usage patterns into debugging artifacts, eval traces, rollout reviews, product requirements, and concrete engineering changes across runtime, model, and UX behavior.

11x.ai | Head of Engineering | Apr. 2023 – Sep. 2023

Fastest-growing UK AI startup from 0 to $1M ARR in under six months. TechCrunch coverage.

- Scaled the codebase and engineering organization behind Alice as 11x.ai grew from 0 to $1M ARR in under six months.
- Built and scaled LLM-backed outbound automation systems that processed millions of email chains and billions of tokens across customer-facing workflows.
- Worked directly with founders, GTM, product, and engineering to turn rapidly changing customer needs into production systems, operating rhythms, and roadmap decisions.

Papaya | Technical Lead | Feb. 2022 – Mar. 2023

Marketplace and fleet platform helping electrify commercial and last-mile logistics. TechCrunch coverage.

- Led delivery of a production fleet management platform in Next.js, React Query, Tailwind CSS, and TypeScript.
- Modernized the Ruby on Rails backend by introducing GraphQL APIs, improving iteration across web, operations, and marketplace workflows.
- Established development cadences, product intake, and stakeholder feedback loops that improved execution across engineering and product.

Weezy (acquired by Getir) | Senior Software Engineer | Aug. 2021 – Jan. 2022

Rapid grocery delivery startup building warehouse, ordering, and customer systems.

- Took over the core ordering and processing monolith in Ruby on Rails, owning day-to-day delivery and forward planning during a high-growth phase.
- Built warehouse operations systems and supporting AWS and Heroku infrastructure with a focus on scale, availability, incident response, and security.

finleap connect | Senior Software Engineer | Nov. 2020 – Aug. 2021

White-label fintech platform powering digital banking products and apps.

- Built GraphQL APIs on NestJS and cross-platform clients in React Native, Apollo, Node.js, and TypeScript for a multi-tenant fintech platform.
- Led alerting and observability improvements while supporting migration from ECS to Amazon Lambda.
- Owned client-specific delivery across the stack, from GraphQL resolvers and mutations to customer-facing mobile screens.

IoT CONNCTD GmbH | Full Stack Engineer | Jul. 2019 – Nov. 2020
Cloud IoT platform unifying device protocols behind a GraphQL API.

- Built frontend interfaces, SDKs, and internal tooling for a cloud IoT platform exposing a unified GraphQL API across multiple device protocols.
- Designed authentication, update, and upstream synchronization flows using TypeScript, React, MobX, and Apollo GraphQL.

SELECTED AI, STARTUP & DEVELOPER TOOLING SIGNALS

Cosine AI Lab: Tech.eu on Cosine: benchmark-leading coding models, agent infrastructure, 38+ language support, and sovereign deployment.
Agent Systems: Why we built our own harness; Cosine Swarm; Secure AI coding agents for enterprise.
OSS & SDKs: codex-web-sdk; claude-agent-sdk-rust; claude-agent-sdk-go; OpenAI-compatible middleware; 100+ merged public GitHub PRs.
Startup Ecosystem: AWS Startups - The Future Of Work; Product Hunt launches; Devpost portfolio.

EDUCATION

Birmingham City University | Bachelor's Degree, Computer Science with Year in Industry | 2015 – 2019

AREAS OF EXPERTISE

AI Deployment & Startups: Startup CTO/Head of Engineering experience, founder support, technical discovery, customer workflows, GTM/product feedback
Agents & AI Applications: OpenAI, Anthropic, Gemini, prompts, tool use, context management, subagents, memory, browser use, MCP/LSP
Evaluations & Model Feedback: RL-style coding gyms, eval design, replay, traces, test/LSP signals, failure modes, model behavior analysis
Software Engineering: TypeScript, Node.js, React, Next.js, GraphQL, Go, Python, APIs, SDKs, distributed systems, observability
Deployment & Trust: Kubernetes, gVisor/runsc, secure execution, private VPC, on-prem, air-gapped deployments, auditability